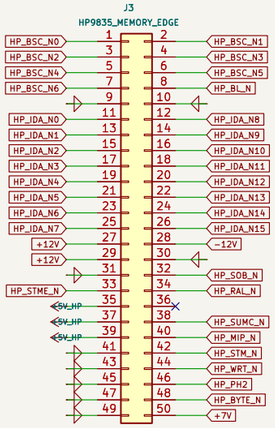
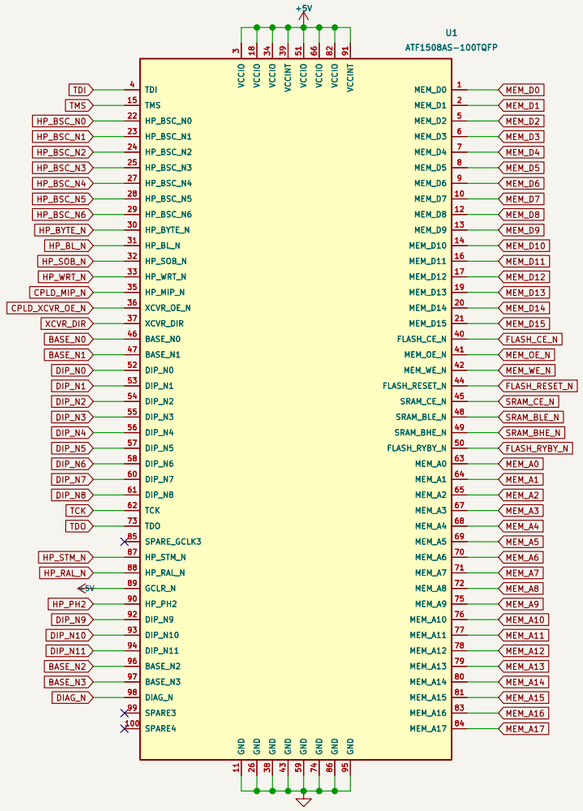
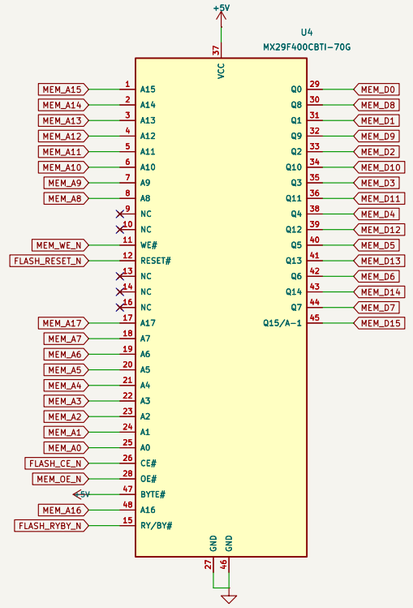
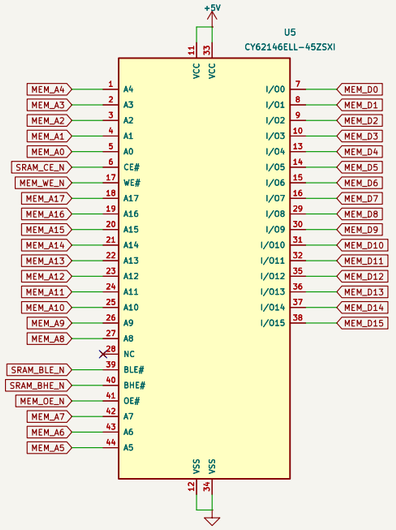
HP memory interface

ATF1508AS TQFP100 — physical pins retained

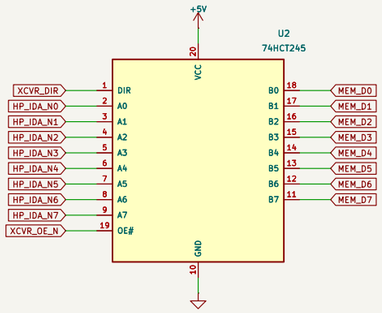


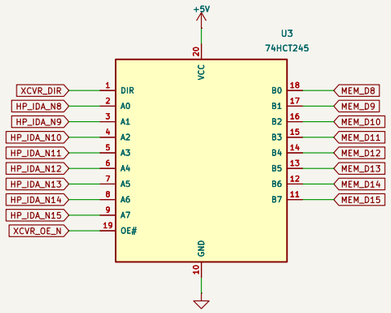






Flash: x16 operation; image payload is complemented.
Shared OE/WE; separate Flash and SRAM CE.
Hardware isolation sheet controls XCVR_OE_N and HP_MIP_N.





Raw HP +5V is isolated from local +5V.
U1 open-collector MIP request is buffered by U13.
No new CPLD logic resources are required.

Configuration and safe idle defaults

File: configuration.kicad_sch

USB / external JTAG assembly options

File: usb_programming.kicad_sch

Power sources and HP bus isolation

File: power_isolation.kicad_sch

Local bypass and bulk capacitance

File: decoupling.kicad_sch

Hardware fit/timing and bench validation required
Assembly: USB or external JTAG; see population CSV

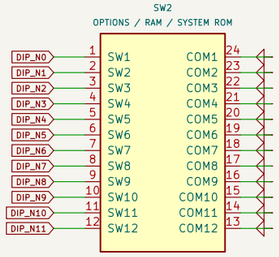
HP 9835 unified memory

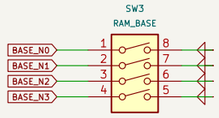
Sheet: /
File: hp9835_unified_ram_rom_board.kicad_sch

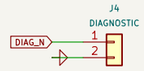
Title: HP 9835 — memory bus and CPLD

Size: A3	Date: 2026-11-07	**Rev: G**
KiCad E.D.A. 10.0.5		Id: 1/5

Configuration switches — open = HIGH; ON/short = LOW



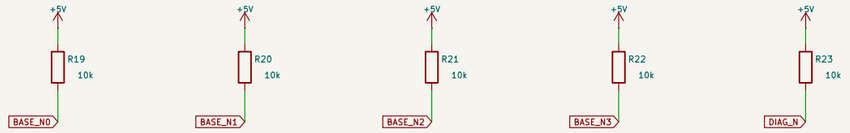




SW2 poles 1–9: option ROMs
Pole 10: reserved A/B selector
Pole 11: RAM enable; pole 12: system ROM
J4: short to enable diagnostic ROM/scratch RAM

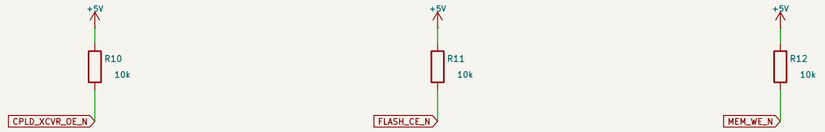
External configuration pull-ups (keepers disabled)

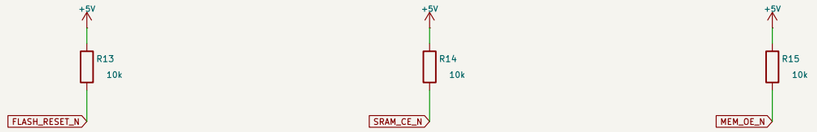




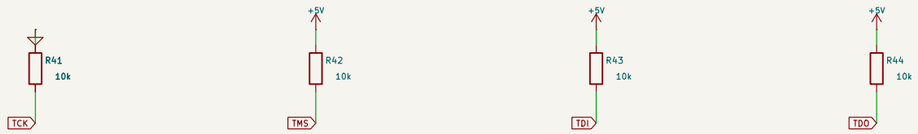


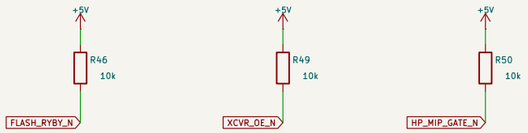
Safe idle defaults





Common JTAG biasing — fitted in BOTH assemblies





Use the base/diagnostic selection before boot.
Configuration HDL and CPLD pad assignments are retained.
All local power bypassing is on sheet 5.

Hardware fit/timing and bench validation required
Assembly: USB or external JTAG; see population CSV
HP 9835 unified memory

Sheet: /Configuration and safe idle defaults/
File: configuration.kicad_sch

Title: Configuration and safe idle defaults

Size: A3	Date: 2026-11-07	**Rev: G**
KiCad E.D.A. 10.0.5		Id: 2/5

ORDER OPTIONS: USB = FT232H circuit, no J2; HEADER = J2, omit USB-only parts

J2
ATF15xx JTAG – THT
TCK
TDO
TMS
TDI
TCK
TDO
TMS
NC
TDI
GND
VCCT
NC
NC
GND
+5V

Keyed 2x5 IDC, 2.54mm, through-hole.
Pin 4 = target supply OUT to cable.
J2 only in HEADER assembly; no FTDI fitted.

J6
USB_B_PROGRAM
VBUS
D+
D-
GND
Shield
VBUS
USB_DP
USB_DM

USB 2.0 device connector; shield bonded to board GND.
Route D+/D- together as a 90-ohm differential pair.
D4 provides low-capacitance USB ESD protection.

VBUS
D4
USBLC6-2SC6
USB_DM
USB_DP
USB_DM
USB_DP

U6
FT232HL
+3V3
VPHY
VPLL
VCCIO
VCCIO
VCCIO
VREGIN
VCCD
VCCCORE
VCCA
DM
DP
RESET
REF
EECS
EECLK
EEDATA
XCSI
XCSO
TEST
AGND
GND
ADBUS0
ADBUS1
ADBUS2
ADBUS3
ADBUS4
ADBUS5
ADBUS6
ADBUS7
ACBUS0
ACBUS1
ACBUS2
ACBUS3
ACBUS4
ACBUS5
ACBUS6
ACBUS7
ACBUS8
ACBUS9
FTDI_TCK
FTDI_TDI
FTDI_TDO
FTDI_TMS
FTDI_JTAG_DISABLE
FTDI_SIWU_N
USB_PWREN_N
USB_DM
USB_DP
FTDI_RESET_N
FTDI_REF
EE_CS
EE_CLK
EE_DATA
FTDI_OSCI
FTDI_OSCO

U8A
74LVC125APW
FTDI_TCK
TCK
USB_JTAG_OE_N

U8B
74LVC125APW
FTDI_TDI
TDI
USB_JTAG_OE_N

U8C
74LVC125APW
FTDI_TMS
TMS
USB_JTAG_OE_N

U8D
74LVC125APW
TDO
FTDI_TDO
USB_JTAG_OE_N

U15
74LVC1G32
+3V3
USB_JTAG_POWER_OFF
FTDI_JTAG_DISABLE
USB_JTAG_OE_N

R24
12k 1%
FTDI_REF

R25
10k
+3V3
FTDI_RESET_N

R26
10k
+3V3
USB_PWREN_N

R45
10k
+3V3
FTDI_SIWU_N

R47
10k
+3V3
FTDI_JTAG_DISABLE

Y1
12MHz / CL=18pF / 30ppm
FTDI_OSCI
FTDI_OSCO

C9
27p
FTDI_OSCI

C10
27p
FTDI_OSCO

U7
93LC56B / x16
+3V3
CS
SCLK
DI
DO
VCC
GND
EE_CS
EE_CLK
EE_DATA
EE_DO

R27
2k2
EE_DATA
EE_DO

R28
10k
+3V3
EE_DO

R29
10k
EE_CS

U8E
74LVC125APW
+3V3
VCC
GND

Nexperia 74LVC125A: partial-power-down isolation.
Enable requires configuration, target supply valid,
AND host MPSSE ready (ACBUS1 LOW).
UART/reset default is disabled by R47.

EEPROM: bus powered, 500mA: ACBUS8 = PWREN#; ACBUS1 = TriState.
Host: select MPSSE, then drive ACBUS1/GPIOH1 LOW to enable JTAG.
Blank EEPROM: target power remains OFF (R26). See tools/openocd_ft232h.cfg.

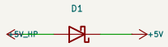
Hardware fit/timing and bench validation required
Assembly: USB or external JTAG; see population CSV
HP 9835 unified memory

Sheet: /USB / external JTAG assembly options/
File: usb_programming.kicad_sch

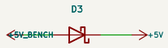
Title: USB / external JTAG assembly options

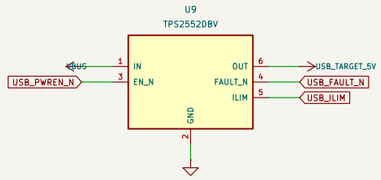
Size: A3	Date: 2026-11-07	**Rev: G**
KiCad E.D.A. 10.0.5		Id: 3/5

Schottky source ORing — all cathodes feed local +5V



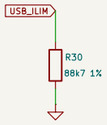


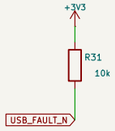


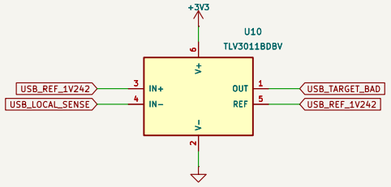


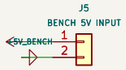




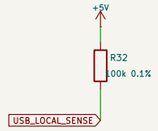


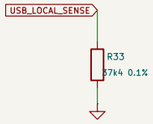


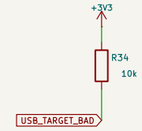




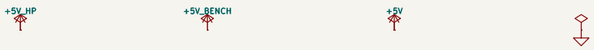
USB target switch: EN# from FT232H PWREN#.
R30 sets target current limit ~299mA nominal.
262–342mA across tolerance (TI datasheet).
USB-only logic enumerates before target power.
Suspend removes the USB target source.

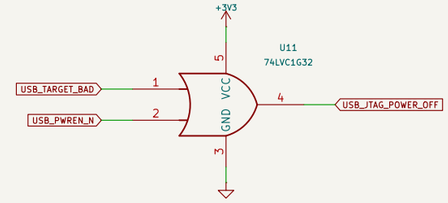


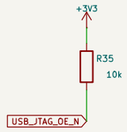




J5 supplies HEADER assembly on the bench.
+5V_HP is raw edge-connector power.
Never bridge raw HP/USB sources to local +5V.
Keep the HP off during installed-card programming.

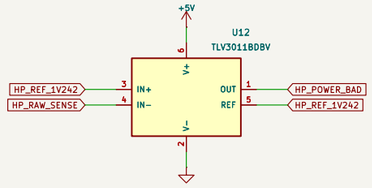


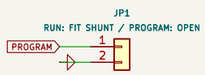


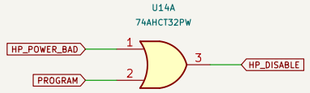


U13 supply is raw HP +5V; Ioff releases MIP when HP is off.
No USB-fed pull-up is connected to the HP MIP bus.
JP1 OPEN forces isolation independently of U1 configuration.

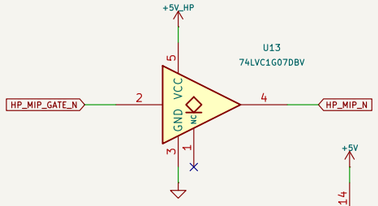
USB JTAG lockout below ~4.56V local supply.
Requires industrial ATF1508AS and valid 5V source:
Schottky ORing does not regulate voltage.

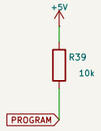


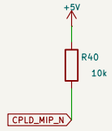


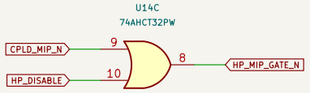


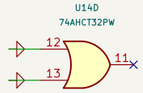


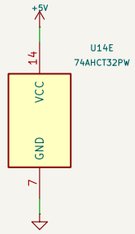


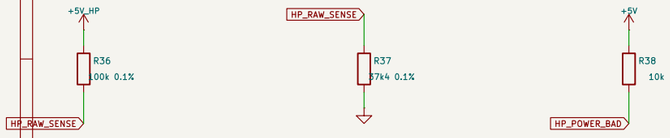












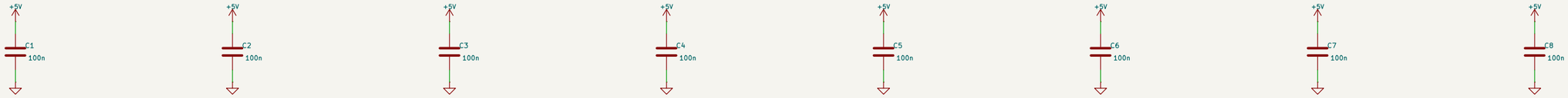
Hardware fit/timing and bench validation required
Assembly: USB or external JTAG; see population CSV
HP 9835 unified memory

Sheet: /Power sources and HP bus isolation/
File: power_isolation.kicad_sch

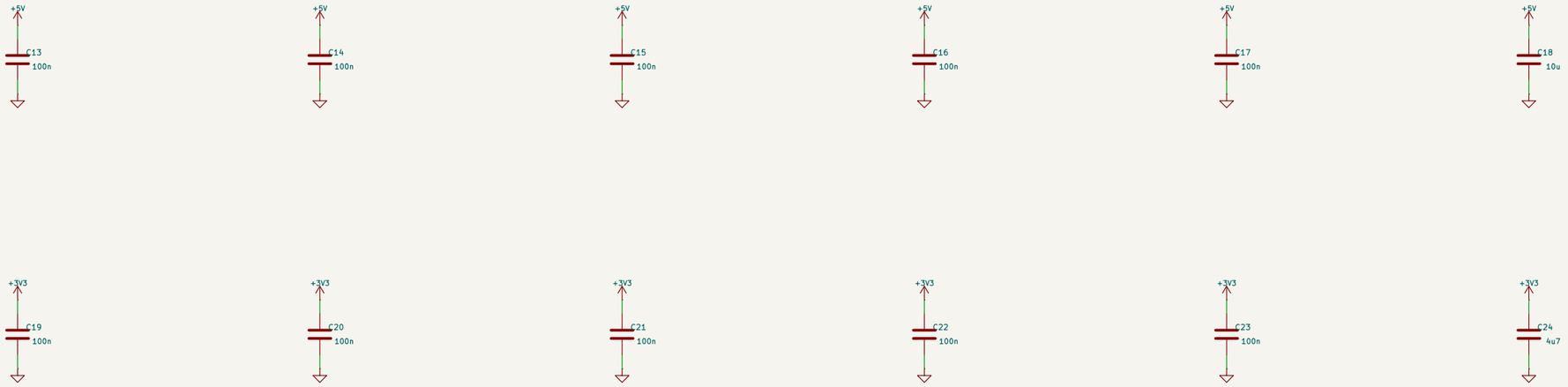
Title: Power sources and HP bus isolation

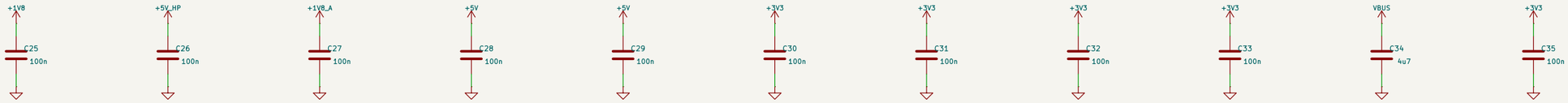
Size: A3	Date: 2026-11-07	**Rev: G**
KiCad E.D.A. 10.0.5		Id: 4/5

U1 — one local capacitor for every VCCINT/VCCIO supply group



U2/U3/U4/U5 and local bulk — fitted in BOTH assemblies





C13 U2; C14 U3; C15 U4; C16/C17 U5. C28 U12; C29 U14.
C19–C23, C24: USB 3.3V groups/bulk. C25/C27: SEPARATE FT232H 1.8V outputs; C34: USB input bulk.
C26 U13; C30 U10; C31 U11; C32 U8; C33 U7; C35 U15.
Keep USB-only bypass parts unpopulated in HEADER assembly.

Hardware fit/timing and bench validation required
Assembly: USB or external JTAG; see population CSV
HP 9835 unified memory

Sheet: /Local bypass and bulk capacitance/
File: decoupling.kicad_sch

Title: Local bypass and bulk capacitance

Size: A3	Date: 2026-11-07	**Rev: G**
KiCad E.D.A. 10.0.5		Id: 5/5